# Project Platinum Shield: Task 1 Report

Proof-of-Concept Firewall Implementation

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## 1 Executive Summary

This report documents the successful deployment of the initial Proof-of-Concept (PoC) firewall environment for Project Platinum Shield. The objective was to establish a secure, routed network topology where a central FreeBSD gateway (VM2) controls traffic flow between a client (VM1) and a target server (VM3). The implementation utilizes FreeBSD's `pf` (Packet Filter) framework to enforce a strict "deny-all" policy, permitting only HTTP traffic on Port 80 while blocking unauthorized protocols such as SSH.

## 2 System Architecture

The experimental sandbox consists of three virtual machines arranged in a linear topology. VM2 acts as the central router and L3 firewall, bridging two distinct subnets.

- **VM1 (Attacker/Client):**
  - **Interface:** 192.168.1.10/24 (Subnet A)
  - **Role:** Simulates external traffic probing the network.
- **VM2 (The Platinum Shield Firewall):**
  - **OS:** FreeBSD 13.4
  - **Interface 1 (em0):** 192.168.1.1 (Gateway for Subnet A)
  - **Interface 2 (em1):** 192.168.2.1 (Gateway for Subnet B)
  - **Role:** Performs IP forwarding and packet filtering via `pf`.
- **VM3 (Target Server):**
  - **Interface:** 192.168.2.20/24 (Subnet B)
  - **Role:** Hosts an Apache HTTP server on Port 80.

## 3 Implementation Steps & Verification

### 3.1 Network Configuration (Requirement A)

**Objective:** Assign private IP addresses and ensure local connectivity within subnets. VM1 was assigned `192.168.1.10`, and VM2 was configured as the gateway with IPs `192.168.1.1` and `192.168.2.1`.

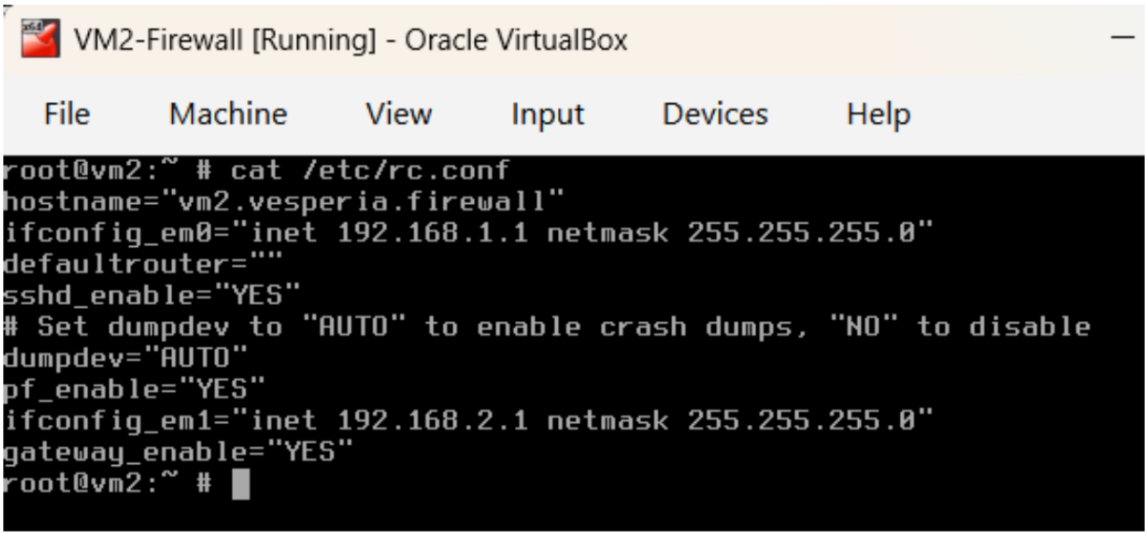


Figure 1: Interface configuration for VM1 (top) and VM2 (bottom), confirming correct subnet assignment.

### 3.2 IP Routing & Traffic Forwarding (Requirement B)

**Objective:** Enable Layer 3 forwarding on VM2 to allow traffic to pass between Subnet A and Subnet B.
**Verification:** A `traceroute` was performed from VM1 to VM3. The output confirms that packets are successfully routed through VM2 (`192.168.1.1`) to reach the target.

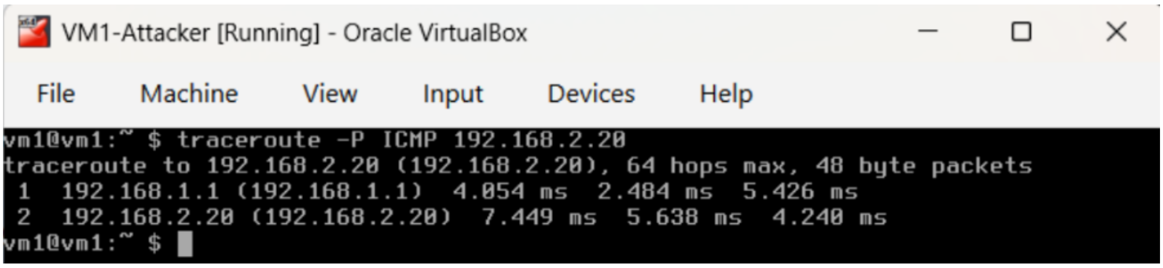


Figure 2: Verification of routing: VM1 successfully traces the route through the gateway.

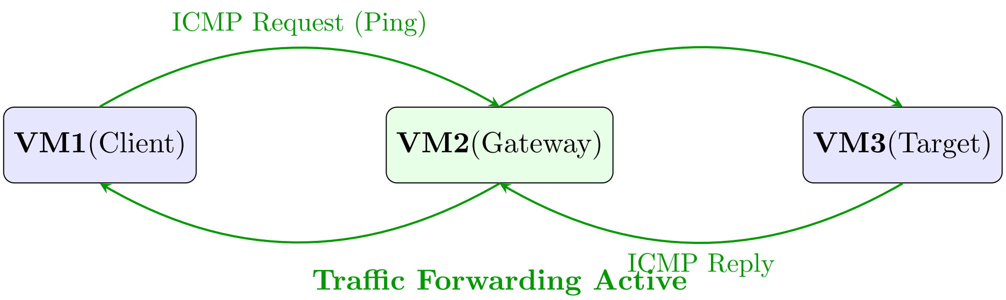


Figure 3: Traffic flow verification: Before firewall activation, ICMP/Traceroute packets are successfully forwarded between subnets.

### 3.3 Firewall Rule Implementation (Requirement D)

**Objective:** Configure `pf` on VM2 to restrict access solely to Port 80 (HTTP). All other traffic, including SSH (Port 22) and ICMP (Ping), must be blocked.
**Configuration (/etc/pf.conf):** The ruleset enforces a default deny policy, explicitly allowing only TCP traffic on Port 80 from the client to the server.

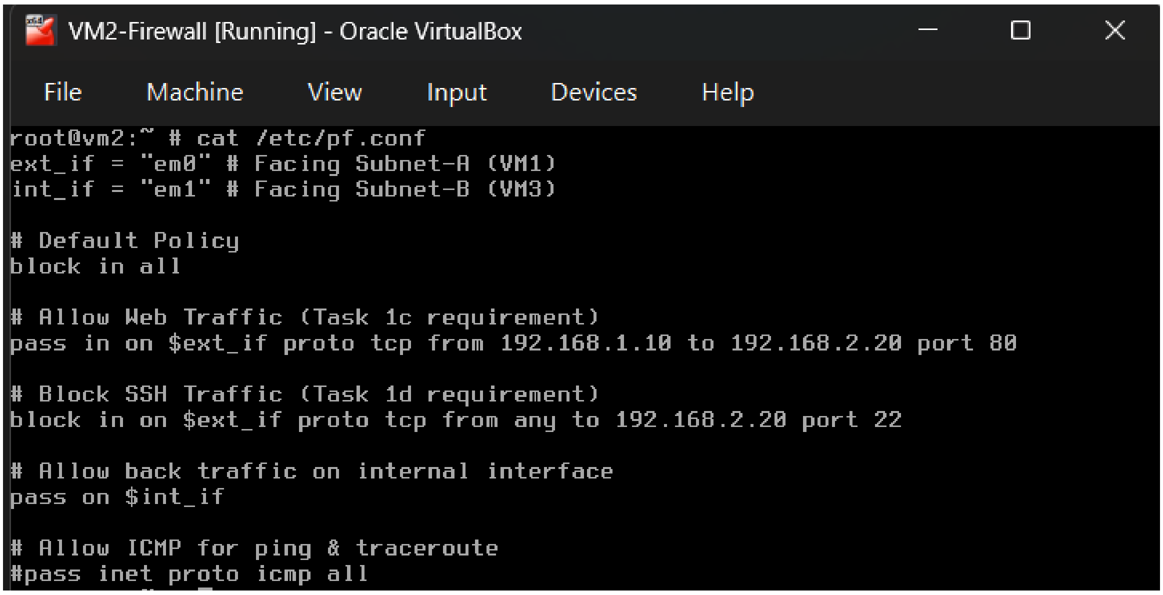


Figure 4: The active `/etc/rc.conf` (top) and firewall rules in `/etc/pf.conf` (bottom) on VM2.

**Verification:** The firewall functionality was verified by attempting to connect via SSH from VM1 to VM3. While the SSH service is running on VM3, the connection from VM1 times out, proving the firewall is blocking Port 22.


Figure 5: Proof of Blocking: SSH is active on VM3 (top), but the connection from VM1 is dropped by the firewall (bottom).

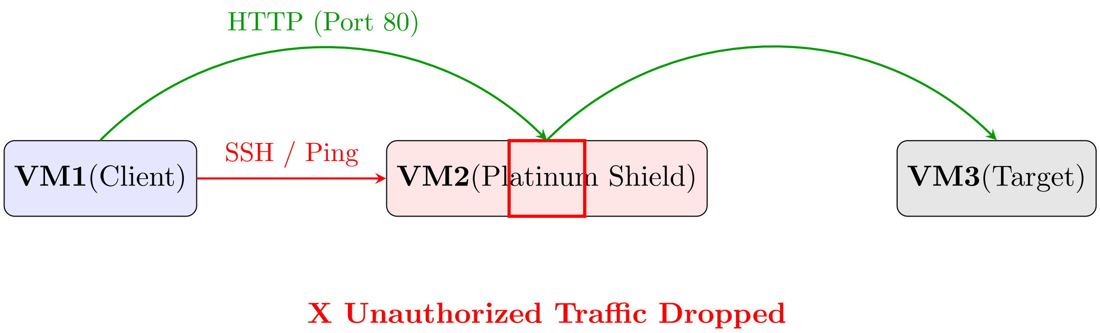


Figure 6: Firewall enforcement: TCP Port 80 is permitted, while SSH and ICMP attempts are terminated at the gateway.

# 4 Conclusion

The Task 1 objectives for Project Platinum Shield have been fully met. The FreeBSD 13.4 firewall successfully routes authorized traffic while strictly enforcing the "deny-all" default policy. This environment is now ready for the development of the custom kernel module required in Task 2.